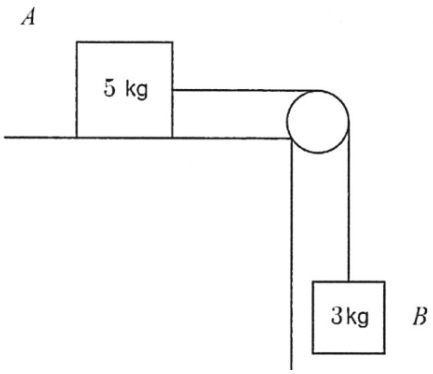
Forces: Connected Particles

Questions 1–8

Question 1. The following questions use $g = 9.8$ **ms**$^{-2}$ Two particles, A and B of mass 5 kg and 3 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.

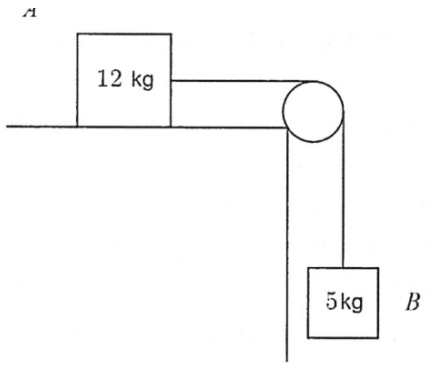


The coefficient of friction between A and the table is 0.25.

a) Find the acceleration of the system.

b) Find the tension in the string.

Question 2. Two particles, A and B of mass 12 kg and 5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.



The two particles are accelerating at 0.02 ms^{-2}. The system is released from rest.
Find

a) the tension in the string.

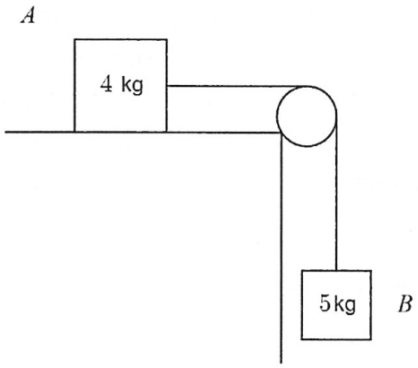
b) the coefficient of friction between A and the table.

Particle B is 0.5 m from the ground. Assuming that particle A does not reach the pulley,

c) find the time it takes for B to reach the ground.

d) state the distance A travels in this time.

Question 3. Two particles, A and B of mass 4 kg and 5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.



The coefficient of friction between A and the table is 0.4

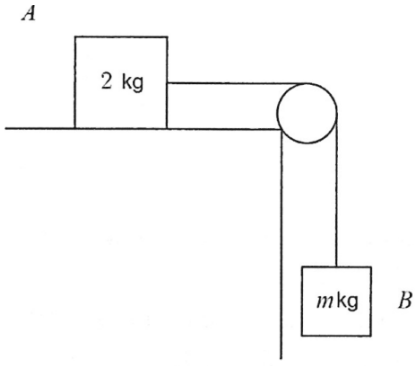
a) Find the acceleration of the system.

b) Find the tension in the string.

The system is released from rest. It takes 1.05 seconds for Particle B to reach the ground and does not reach the pulley.

c) Determine the initial height of particle B above the ground.

Question 4. The following questions use exact values of g Two particles, A and B of mass 2 kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string.

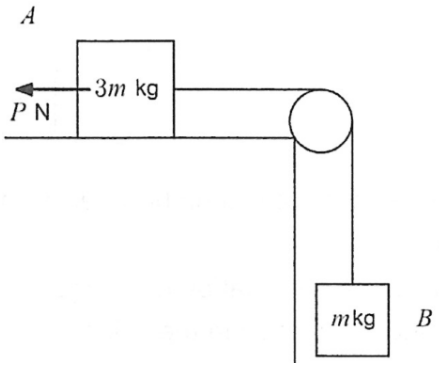


The coefficient of friction between A and the table is 0.3

a) Find the acceleration of the system in terms of m and g

b) Find the minimum value of m such that Particle B is moving.

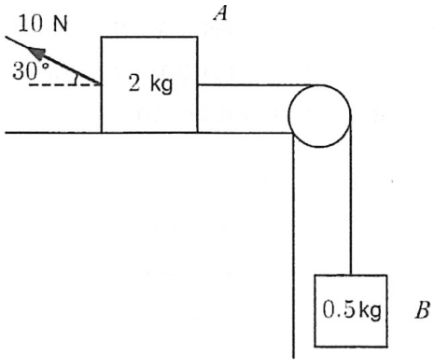
Question 5. The following questions use $g = 9.8$ **ms**$^{-2}$ Two particles A and B of mass $3m$ kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. The coefficient of friction between A and the table is μ. Particle B hangs freely with the string taut. Particle A is pulled away from the pulley with horizontal force P N with acceleration a ms^{-2}.



Show that

$$P = (29.4\mu + 4a + 9.8)m$$

Question 6. Two particles, A and B of mass 2 kg and 0.5 kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. Particle B hangs freely at the end of the string. Particle A is being pulled by a force of 10 N at an angle 30° to the horizontal away from the pulley.



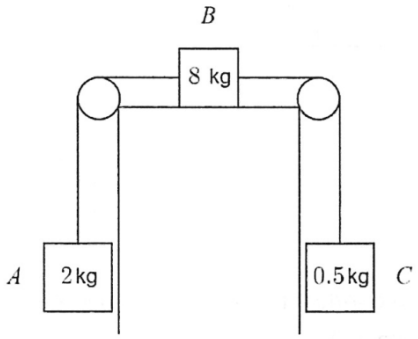
The coefficient of friction between A and the table is 0.2. Find

a) the frictional force acting on particle A
b) the acceleration of the system.
c) the tension in the string.

Particle B is swapped for particle C of mass m kg.

d) Find the maximum value of m such that C accelerate upwards.

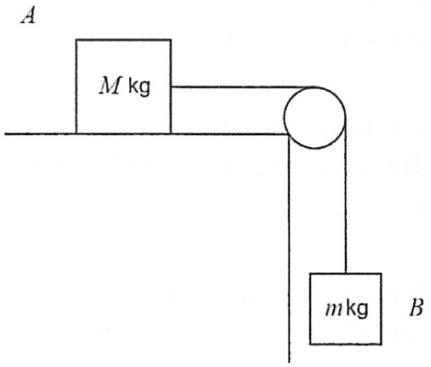
Question 7. Three boxes, A, B and C have masses 2 kg, 8 kg and 0.5 kg respectively. Box A is hanging freely from a string, which passes over a small, fixed, smooth pulley and connects to box B. Box B is on a rough horizontal table. Box C is hanging freely from a string, which passes over a small, fixed, smooth pulley and connects to the other side of box B.



The coefficient of friction between box B and the table is 0.1. The system is moving.

a) Find the tension in the strings.
b) Find the acceleration of the system and describe the motion of the boxes.
c) State the modelling assumptions you have used.

Question 8. The following questions use exact values of g Two particles A and B of mass M kg and m kg respectively, are attached by a light inextensible string over a small smooth peg. Particle A rests on a rough table. The coefficient of friction between A and the table is μ. Particle B hangs freely with the string taut.



The system is moving. Show that

$$m \geq \mu M$$